

Practical Problem-03

Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 void display_process_info(const char *stage)
8 {
9     printf("\n--- %s ---\n", stage);
10    printf("PID : %d\n", getpid());
11    printf("PPID : %d\n", getppid());
12 }
13
14 int main()
15 {
16     pid_t pid;
17
18     printf("Starting Parent Process\n");
19
20     display_process_info("Before fork()");
21
22     pid = fork();
23
24     if (pid < 0)
25     {
26         perror("fork failed");
27         return 1;
28     }
29
30     else if (pid == 0)
31     {
32         // Child process
33         display_process_info("Child Process - Running");
34
35         printf("Child is now sleeping (Waiting state)...\n");
36
37         sleep(10);
38     }
```

0

```
display_process_info("Child Process - Running Again");

printf("Child process is terminating...\n");

exit(0);
}

else
{
    // Parent process
    display_process_info("Parent Process - Running");

    printf("Child PID: %d\n", pid);

    printf("Parent is waiting for child to finish...\n");

    wait(NULL);

    display_process_info("Parent After wait()");

    printf("Child has terminated.\n");
}

return 0;
```

Output:

```
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ps
  PID TTY          TIME CMD
   313 pts/0    00:00:00 bash
  1989 pts/0    00:00:00 ps
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ top
top - 09:29:18 up 1:29, 1 user, load average: 0.07, 0.04, 0.00
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15712.1 total, 15107.5 free, 630.2 used, 172.8 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 15081.9 avail Mem
```

```
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ./Practical3
Starting Parent Process

--- Before fork() ---
PID : 1947
PPID : 313

--- Parent Process - Running ---
PID : 1947
PPID : 313
Child PID: 1948
Parent is waiting for child to finish...

--- Child Process - Running ---
PID : 1948
PPID : 1947
Child is now sleeping (Waiting state)...

--- Child Process - Running Again ---
PID : 1948
PPID : 1947
Child process is terminating...

--- Parent After wait() ---
PID : 1947
PPID : 313
Child has terminated.
```

```
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ /proc
-bash: /proc: Is a directory
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ls /proc
1      2023  410  846  consoles      filesystems    kmsg           mounts        slabinfo      uptime
106    234   48   850  cpuinfo       fs             kpagecgroup    mpt           softirqs      version
107    238   8    858  crypto        interrupts     kpagecount     mtrr          stat          vmallocinfo
108    275   80   89   devices       iomem          kpageflags     net           swaps         vmstat
112    311   818  acpi   diskstats     ioports        latency_stats  pagetypeinfo  sys           zoneinfo
1162   312   822  buddyinfo  dma            irq            loadavg        partitions    sysrq-trigger
161    313   829  bus       driver         kallsyms       locks          pressure      sysvipc
185    314   830  cgroups   dynamic_debug kcore          meminfo        schedstat     thread-self
198    380   835  cmdline   execdomains   key-users      misc           scsi          timer_list
2      384   840  config.gz  fb             keys           modules        self          tty
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ gedit Practical3.c

** (gedit:2136): WARNING **: 09:44:10.833: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:2136): WARNING **: 09:44:10.833: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$
```